

Solve multi-step proportion problems

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance. Copy or complete the first step.

2. A length of 7 cm is enlarged by scale factor 3.5. What is the new length?

3. A recipe for 6 uses 450 g flour. How much flour is needed for 14?

4. Miro says: Miro uses the scale factor on only one corresponding quantity. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance. Copy or complete the first step.
Answer 52 km.
2. A length of 7 cm is enlarged by scale factor 3.5. What is the new length?
Answer 24.5 cm.
3. A recipe for 6 uses 450 g flour. How much flour is needed for 14?
Answer 1,050 g.
4. Miro says: Miro uses the scale factor on only one corresponding quantity. Is this correct? Explain.
Answer Apply the same multiplicative relationship to every corresponding measure or ratio part.

Solve multi-step proportion problems

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.

6. Check this result using an inverse, estimate or known fact: A recipe for 6 uses 450 g flour. How much flour is needed for 14?

2. A length of 7 cm is enlarged by scale factor 3.5. What is the new length?

7. Prove or explain your reasoning: In the ratio 4:7, the smaller part is 20. What is the larger part?

3. A recipe for 6 uses 450 g flour. How much flour is needed for 14?

8. Identify and correct this misconception: Miro uses the scale factor on only one corresponding quantity.

4. Solve this independently and show every stage: A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: A length of 7 cm is enlarged by scale factor 3.5. What is the new length?

10. Write and solve a similar question to: A recipe for 6 uses 450 g flour. How much flour is needed for 14?

Teacher answers and checking prompts

1. A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.
Answer 52 km.
2. A length of 7 cm is enlarged by scale factor 3.5. What is the new length?
Answer 24.5 cm.
3. A recipe for 6 uses 450 g flour. How much flour is needed for 14?
Answer 1,050 g.
4. Solve this independently and show every stage: A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.
Answer 52 km.
5. Use a second method or representation for: A length of 7 cm is enlarged by scale factor 3.5. What is the new length?
Answer Expected result: 24.5 cm. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: A recipe for 6 uses 450 g flour. How much flour is needed for 14?
Answer Expected result: 1,050 g. A valid check must be shown.
7. Prove or explain your reasoning: In the ratio 4:7, the smaller part is 20. What is the larger part?
Answer 35. The explanation must show why the method works.
8. Identify and correct this misconception: Miro uses the scale factor on only one corresponding quantity.
Answer Apply the same multiplicative relationship to every corresponding measure or ratio part.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: A recipe for 6 uses 450 g flour. How much flour is needed for 14?
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Solve multi-step proportion problems

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: In the ratio 4:7, the smaller part is 20. What is the larger part?

6. Create a new example that tests the same idea as: A recipe for 6 uses 450 g flour. How much flour is needed for 14?

2. Prove the answer to this question, rather than only calculating it: A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: A length of 7 cm is enlarged by scale factor 3.5. What is the new length?

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 1,050 g.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro uses the scale factor on only one corresponding quantity.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: In the ratio 4:7, the smaller part is 20. What is the larger part?
Answer 35. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: A map scale is 1 cm to 8 km. Two towns are 6.5 cm apart. Find the real distance.
Answer Expected result: 52 km. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: A length of 7 cm is enlarged by scale factor 3.5. What is the new length?
Answer Expected result: 24.5 cm. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 1,050 g.
Answer One valid reconstruction is: A recipe for 6 uses 450 g flour. How much flour is needed for 14? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro uses the scale factor on only one corresponding quantity.
Answer Apply the same multiplicative relationship to every corresponding measure or ratio part.
6. Create a new example that tests the same idea as: A recipe for 6 uses 450 g flour. How much flour is needed for 14?
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.